

VALERY LAVROV

Independent Product Developer

Lod, Israel
lavrov-dev.com

+972 543 328 803
linkedin.com/in/valery-lavrov

valerynz@hotmail.com
github.com/ValeryKG

PROFESSIONAL SUMMARY

Six production apps. Real problems identified. Real people using them daily. Every idea, every product decision, every architecture choice — mine. I conceive products from direct exposure to real problems, design the solution, and own the full lifecycle from data architecture through legal compliance and deployment.

I build with a repeatable workflow that improves with every project. Every bug that matters becomes a rule. Every architecture decision is documented. The next build starts with everything the previous ones learned — faster delivery, fewer repeated mistakes, codebases that stay understandable a year after launch.

Background spans Java enterprise development (Oracle Certified, 93%/98%), React/TypeScript/Firebase, web agency ownership, and 20 years as a professional basketball coach — where I applied data analytics and systematic planning to a field most people treat as pure intuition. Fluent English, native Russian, conversational Hebrew.

FEATURED PROJECT

TimeClock GPS — Multi-Tenant Workforce Management SaaS

React 19 · TypeScript · Firebase · bcrypt · GPS/Geolocation · React i18next · date-fns · jsPDF · xlsx · ~18,600 lines · 31 calendar days, evenings only

- Multi-tenant SaaS — any organization deploys independently with their own roles, sites, and approval configuration
- GPS-verified clock-in/out for multi-site organizations — 7 user roles, 7 dashboards including Accountant and Observer
- Device-level fraud prevention: UUID fingerprinting in IndexedDB, pending approval queue, per-device revocation by manager
- Observer role: neutral third-party read-only access — eliminates payroll disputes through shared, uneditable truth
- Five GPS approval modes per site: auto-approve, GPS check, hours range, expected-hours, strict shifts
- Time record review workflow with auto-flagging of anomalies — Accountant dashboard with payroll summaries
- PDF and Excel payroll export — 4 languages, Hebrew-first, full RTL support
- Offline-first: clock-in/out queues locally and syncs automatically on reconnect via IndexedDB

ALL PROJECTS (2022 – 2026)

StockPilot — Facilities Inventory & Maintenance Platform · inventory-e5daa.web.app

React 19 · TypeScript · Firebase · Zod · Recharts · ~11,500 lines · ~3 weeks, evenings · v1.1.7

Multi-role inventory and maintenance platform in active daily use by the facilities team at Palo Alto Networks. Identified the problem directly from working in the same environment — spreadsheets, WhatsApp, and no shared system. Built the replacement. Separates consumables from assets: assets carry serial number, assignment, and repair status. Maintenance ticket system with full lifecycle (open → in progress → escalated → resolved) — every status change auto-logs to an immutable timeline enforced at the Firestore rule level. Asset inspection logging with one-tap check-in and service history. PDF export per ticket and batch Excel/PDF reports filtered by status and date range. Automatic low-stock alerts. Folder-level ACL. 12 Architecture Decision Records.

WishBasket — Family Home Management · wishbasket.app

React 19 · TypeScript · Firebase · i18next · jsPDF · PWA · ~9,700 lines · ~2 weeks, evenings

Family inventory and accountability app built around one idea: any member requests an item, any other member claims it and owns the outcome — name recorded, result logged permanently and immutably at the Firestore rule level. Two-stage Request & Promise flow (pending → promised → resolved). Folder-level ACL with zero per-item query overhead. Custom org-defined statuses. Kosher/halal tracking baked into the data model. Shopping Mode overlay. Built-in Help & Feedback system with role-aware user guide. PWA install prompt handled for Android (native) and iOS (manual guide fallback). 3 languages with RTL. Real families in active use.

CourtIQ — Basketball Player Development Platform · court-iq.org

React 18 · TypeScript · Firebase · Tailwind CSS v4 · PWA · Recharts · ~20,000 lines

Sports management platform with three user roles (Player, Coach, Manager). Real-time skill tracking, session logging, scouting reports, scheduling, ROI analytics. 176-skill library. Full legal compliance — ToS, Privacy Policy, consent tracking.

MyHours — Personal Time Tracker

React 18 · TypeScript · Firebase · IndexedDB · jsPDF · xlsx · ~5,700 lines

PWA for freelancers and employees. Multi-workplace GPS auto-detection, hourly rates, overtime multipliers, time rounding (8 settings per workplace). Multi-currency (ILS, USD, EUR, GBP, RUB), PDF/Excel payroll reports. 4 languages with RTL. Offline sync.

Baiti — Building Management Platform · baiti.co.il

Vanilla JavaScript · Firebase (Realtime DB, Auth, Cloud Messaging, Functions) · Tailwind CSS · ~22,000 lines

Apartment building management: payment tracking, tenant admin, voting system, push notifications — no app store required. Built without a JS framework. 30+ iterations from prototype to version 1.0.9. Hebrew, Russian, English with RTL.

HOW I BUILD

40-document cross-project knowledge base — built from real failures across 6 production apps.

Every bug that mattered became a rule. Every architecture decision documented with the problem, the wrong approach considered, and what was chosen and why. A new project starts with everything the previous ones learned — which means the sixth app was built faster and with fewer mistakes than the first, and is still understandable a year later without archaeology.

Includes: a 6-stage decision tree that gates architecture before the first commit, a session protocol that maintains context across AI-assisted development sessions, 12 dated Architecture Decision Records from StockPilot alone, device token and approval queue patterns, PIN + bcrypt auth patterns, security and launch checklists, and domain-specific guides updated from real production bugs.

TECHNICAL SKILLS

Frontend	React 18/19, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Vite
Backend & Cloud	Firebase (Firestore, Auth, Hosting, Functions, Storage), Node.js, Java SE/EE, JSP, Servlets, JDBC, EJB
Databases	Firebase Firestore, Oracle DB, MySQL, SQL, PL/SQL
PWA & Offline	Service Workers, IndexedDB, Web Manifest, offline-first architecture, auto-update systems
APIs & Libraries	Browser Geolocation, Haversine formula, jsPDF, SheetJS (xlsx), Recharts, Zod, Chart.js, date-fns, dnd-kit
Auth & Security	PIN + bcrypt hashing, device fingerprinting, RBAC, Firestore security rules, session management
i18n	React i18next — English, Hebrew (RTL), Arabic (RTL), Russian
Tooling	Git/GitHub, VS Code, Firebase CLI, ESLint, TypeScript strict mode
Legacy	Apache Tomcat, WebLogic, GlassFish, Oracle Application Server, Java EE

WORK HISTORY

Independent Product Developer

2022 – Present

Self-employed · Israel

Conceiving, building, and shipping complete production applications solo — every idea, every product decision, every architecture choice mine. Six live apps across workforce management, inventory control, building management, and sports analytics.

- TimeClock GPS: multi-tenant workforce SaaS, 7 user roles, device-level fraud prevention, 5 GPS approval modes, payroll reporting — 31 days evenings
- StockPilot: facilities inventory and maintenance platform — ticket system with immutable audit trail, asset inspection logging, in active daily use at Palo Alto Networks
- WishBasket: family accountability app with peer Request & Promise system and immutable Firestore log — live at wishbasket.app
- CourtIQ: basketball coaching platform with skill tracking, scouting history, and 176-skill library
- Baiti: building management platform with push notifications and multilingual payment tracking
- MyHours: personal time tracker with GPS, multi-workplace settings, and PDF/Excel payroll export
- All 6 apps multilingual — Hebrew (RTL), Arabic (RTL), Russian, English
- 40-document cross-project knowledge base — architecture decisions, security patterns, deployment protocols — updated from real production failures, applied to every build

Facilities Technician

April 2022 – Present

Palo Alto Networks (formerly CyberArk) · Petah Tikva, Israel

Facilities operations at a global cybersecurity company. Identified gaps in how the maintenance team tracked inventory and issues — built StockPilot as the solution. Now in active daily use by the same team.

Head Basketball Coach

September 2023 – Present

Hapoel Lod · Israel

Leading program transformation through systematic coaching and data-driven analytics. PPP-based shot efficiency analysis, pick-and-roll coverage effectiveness rankings, and position-flexible player development.

Web Developer / Founder

2014 – 2017

Israweb4u · Israel

Web design, development, and digital marketing for Israeli small business clients. Facebook Ads and Google Ads campaign management.

Java Developer

2010 – 2012

Unlimited Capital Ltd · Israel

Designed and built a Java EE enterprise web application. Secure member registration (SHA hashing), warehouse/order processing system, admin dashboard. Stack: Java SE/EE, Servlets, JSP, EJB, JDBC, Oracle DB, WebLogic, JUnit. Concurrent with Oracle certification — 93% Java SE, 98% SQL/PL/SQL.

IT Administrator / Head Basketball Coach

2007 – 2010

YouthTown · Auckland, New Zealand

Dual role: IT network administration and head coaching for youth basketball program.

Computer Technician / Sales

2005 – 2006

GOLF Computer Company Ltd · Israel

Computer Support Specialist

2002 – 2004

SKORNIK Security Ltd · Israel

Business Owner — PC Parts & Software Store

1995 – 1997

Self-employed · Simferopol, Ukraine

Procured and retailed PC hardware, games, and software. First business ownership experience.

EDUCATION & CERTIFICATIONS

Oracle Certified Professional

2010 – 2012

Oracle Space · Israel

- Oracle SQL, PL/SQL, DBA — Score: 98%
- Java SE 6 / J2EE Certified Professional — Score: 93% (passing threshold: 61%)

Licensed Basketball Coach

2013 – 2014

Wingate Institute — Tel Aviv University · Israel

Professional coaching certification. Certificate No. 13-34-0001908.

Master's Degree — Sport Science

1997 – 2003

Tavrian National University · Simferopol, Ukraine

Specialist of Physical Training (coach, instructor, PE). Diploma No. A 388339.

LANGUAGES

English — Fluent · **Russian** — Native · **Hebrew** — Conversational